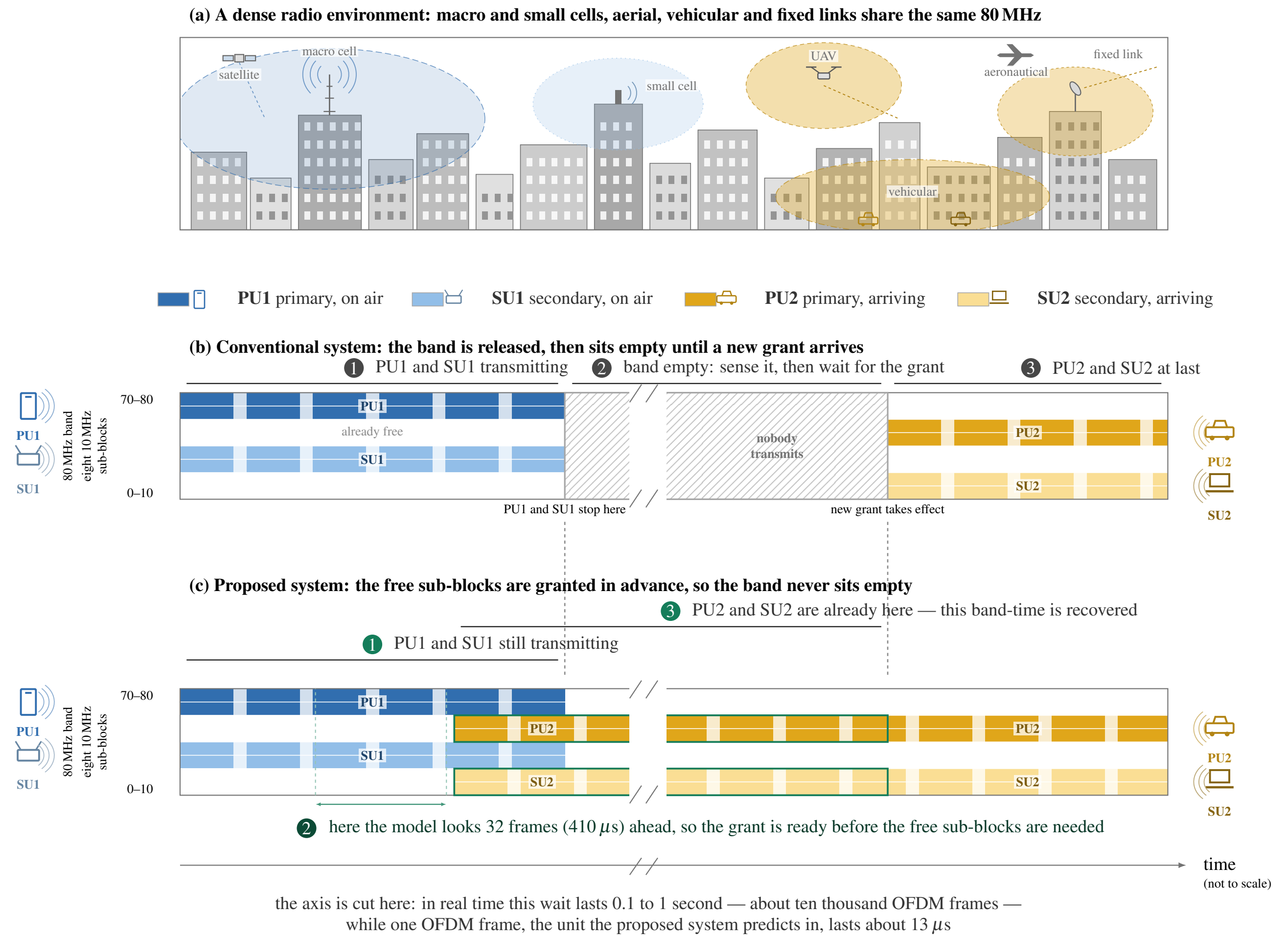
(a) A dense radio environment: macro and small cells, aerial, vehicular and fixed links share the same 80 MHz
satellite
macro cell
small cell
UAV
aeronautical
fixed link
vehicular
PU1 primary, on air
SU1 secondary, on air
PU2 primary, arriving
SU2 secondary, arriving
(b) Conventional system: the band is released, then sits empty until a new grant arrives
1 PU1 and SU1 transmitting
2 band empty: sense it, then wait for the grant
3 PU2 and SU2 at last
PU1
SU1
80 MHz band
eight 10 MHz sub-blocks
70–80
0–10
PU1
already free
SU1
nobody transmits
PU2
SU2
PU2
SU2
PU1 and SU1 stop here
new grant takes effect
(c) Proposed system: the free sub-blocks are granted in advance, so the band never sits empty
3 PU2 and SU2 are already here — this band-time is recovered
1 PU1 and SU1 still transmitting
PU1
SU1
80 MHz band
eight 10 MHz sub-blocks
70–80
0–10
PU1
PU2
SU1
SU2
PU2
SU2
PU2
SU2
2 here the model looks 32 frames (410 μs) ahead, so the grant is ready before the free sub-blocks are needed
time
(not to scale)
the axis is cut here: in real time this wait lasts 0.1 to 1 second — about ten thousand OFDM frames —
while one OFDM frame, the unit the proposed system predicts in, lasts about 13 μs